

Corporate Finance Assignment Documentation

1. Introduction

This project presents a comprehensive financial analysis of NTPC Ltd. and Kalpataru Projects International Ltd. using financial statements, stock market data, bond market information, and Microsoft Excel. The analysis focuses on corporate governance, capital structure, working capital management, cost of capital, and financial decision-making. All calculations and interpretations were performed using an Excel-based financial model following the assignment guidelines.

2. Objectives

- Estimate the equity beta of NTPC using regression analysis.
- Calculate the Cost of Equity using the Capital Asset Pricing Model (CAPM).
- Analyse the company's long-term debt and estimate the Cost of Debt.
- Compute the Weighted Average Cost of Capital (WACC).
- Evaluate the corporate governance practices of Kalpataru Projects International Ltd.
- Analyse the company's working capital management over the last three years.
- Study capital expenditure trends and investment decisions.
- Examine capital structure and financing strategy.
- Estimate WACC, corporate growth rate, Degree of Operating Leverage (DOL), and Degree of Financial Leverage (DFL).

3. Company 1 – NTPC Ltd.: Beta Estimation and Cost of Equity

Three years of weekly closing prices of NTPC and the NIFTY 50 Index were collected and converted into weekly percentage returns. Linear regression was performed to estimate beta. The regression output included beta, t-statistic, p-value, R^2 and number of observations. The estimated beta was then used in the CAPM model to determine the cost of equity and interpret the systematic risk of the company.

4. Bond Analysis and Cost of Debt

The assigned corporate bond was analysed using its ISIN. Coupon rate, face value, maturity date, credit rating, yield to maturity (YTM), and premium/discount status were

examined. Yield spread over the nearest government bond was interpreted in terms of credit risk, liquidity, and market conditions. The YTM was used as the firm's pre-tax cost of debt.

5. Weighted Average Cost of Capital (WACC)

Using the estimated cost of equity, cost of debt, tax rate, and capital structure, the firm's WACC was calculated and interpreted.

6. Company 2 – Corporate Governance Analysis

The governance framework was evaluated using the latest annual report. The study examined promoter ownership, board size, CEO duality, gender diversity, independent directors, executive and non-executive directors, auditor independence, significant litigation, and information asymmetry.

7. Working Capital Analysis

Current assets, current liabilities, net working capital, current ratio, and quick ratio were analysed for the past three years to assess liquidity and short-term financial strength.

8. Capital Expenditure Analysis

Capital expenditure over the last three years was analysed to identify major investments, classify projects, evaluate ROIC, and identify funding sources.

9. Capital Structure Analysis

Quarterly capital structure data was analysed to study debt-equity trends, funding sources, operating cash flows, and the firm's ability to finance capital expenditure and financial obligations.

10. WACC and Growth Rate Estimation

The firm's WACC, sustainable growth rate, DOL, and DFL were estimated and interpreted to assess financial risk and growth potential.

11. Tools and Skills Applied

- Financial Statement Analysis
- Corporate Governance Evaluation
- Regression Analysis
- Beta Estimation
- CAPM
- Bond Valuation
- Cost of Debt Estimation
- Weighted Average Cost of Capital (WACC)
- Working Capital Analysis
- Capital Structure Analysis
- Return on Invested Capital (ROIC)
- Degree of Operating Leverage (DOL)
- Degree of Financial Leverage (DFL)
- Microsoft Excel Financial Modeling

12. Conclusion

This project provided a comprehensive analysis of the financing, investment, and governance decisions of NTPC Ltd. and Kalpataru Projects International Ltd. By integrating regression analysis, bond valuation, corporate governance assessment, working capital evaluation, capital structure analysis, and WACC estimation, the study offered insights into each company's financial health and strategic decision-making. The Excel-based implementation ensured transparency, accuracy, and practical application of key corporate finance concepts.